

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think>

the total size of the file is 200 GB

Carla's normal download speed is 2 GB / minute

to download 200 GB at 2 GB / minute , it would take $200 / 2 = 100$ minutes

but 40 % of the way through the download , which is $0.4 * 200 = 80$ GB , has already been downloaded

so Carla has already spent $80 / 2 = 40$ minutes downloading

then the 20 minutes are spent installing the updates

so the total time spent before the restart is $40 + 20 = 60$ minutes

after the restart , Carla has to download the entire 200 GB again

so the total time it takes to download the file is $60 + 100 = 160$ minutes

</think>< answer > 160 </answer>

